

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester II Examination, 2024-25 (NEP)

PHYSICS**Paper Code: PHYMJ21 / PHYMN21 : Thermal Physics**

Ref. No.: CECNF/N/2024-25/4

Time: $2\frac{1}{2}$ Hours

Maximum Marks: 70

Instructions: Attempt five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer all questions briefly: [20]

- (a) State the zeroth law of thermodynamics.
- (b) Explain Zartman and Ko's experiment for verification of Maxwell's law of distribution of speeds.
- (c) Calculate the mean free path for molecules of hydrogen gas. The radius of the gas molecule is 1.37×10^{-10} m and the density of gas is $3 \times 10^{25} \text{ m}^{-3}$.
- (d) Write photon gas pressure inside a cavity in terms of photon's energy density.
- (e) Draw T-S diagram of a reversible Carnot cycle.
- (f) Calculate the change in entropy when 1.119 kg of water at 100°C is converted into steam at the same temperature. Latent heat of steam is 540 cal/gram.
- (g) Write inversion temperature in terms of real gas constants.
- (h) Write the relation between Gibbs and Helmholtz free energy.
- (i) Explain the triple point of water using a suitable diagram.
- (j) Explain Joule-Thomson effect.

Question 2.

- (a) Explain the Carnot cycle. Show that no irreversible heat engine operating between two reservoirs (source and sink having absolute temperatures T_1 and T_2 respectively) can be more efficient than a reversible engine operating between the same reservoirs. [$8\frac{1}{2}$]
- (b) Calculate the work done in kJ during an isothermal reversible process when 5 moles of ideal gas are expanded so that its volume is doubled at a temperature of 400 K. [4]

Question 3.

- (a) Show that the change in entropy is zero for a reversible cyclic process, but it increases for an irreversible process. [7]
- (b) Write the relation between heat capacities at constant pressure and at constant volume as a function of isothermal compressibility and volume expansivity. Calculate the difference between them for ^4He gas at temperature 6 K and pressure 19.7 atm. [Given: specific volume = $2.64 \times 10^{-2} \text{ m}^3 \text{ kmol}^{-1}$; isothermal compressibility = $9.42 \times 10^{-8} \text{ m}^2 \text{ N}^{-1}$; volume expansivity = $5.35 \times 10^{-2} \text{ K}^{-1}$]. [$5\frac{1}{2}$]

Question 4.

- (a) Derive Clausius-Clapeyron equation using Carnot's cycle. [7½]
- (b) Prove the following relation: [5]

$$T \cdot dS = C_p dT - T \left(\frac{\partial V}{\partial T} \right)_p dP$$

Question 5.

- (a) Derive the Rayleigh-Jeans law. Also explain the UV catastrophe. [8½]
- (b) The sun's radius is 6.96×10^8 m and its surface temperature is 5700 K. Estimate the total power radiated by the Sun. (Stefan's constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$). [4]

Question 6.

- (a) Explain transport phenomena in gases. Derive the expression for viscosity of gases. [7½]
- (b) Write critical gas constants in terms of real gas constants. Find the gas constant b if critical constant V_c is 69.7 cc/mole. [5]

Question 7.

- (a) Derive the following Maxwell's thermodynamic relation: [5]

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

- (b) Explain the principle of regenerative cooling with a schematic diagram. [7½]
